Spring 2016

Marmara University
Faculty of Engineering
Electrical and Electronics Engineering
STAT2056-Probability and Random Variables
Final Exam

1. The independent discrete random variables X and Y are given by

$$X = 2\cos\left(\frac{2\pi\theta}{8}\right), \qquad Y = 2\sin\left(\frac{2\pi\theta}{8}\right)$$

where θ is a discrete uniform random variable in the set $\{0,1,\cdots,7\}$. That is,

$$p_\theta(k) = P[\theta = k] = \begin{cases} 1/8, & k = 0,1,\cdots,7 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find $p_{X,Y}[i,j]$, the joint probability mass function (PMF) of X and Y. [10 pts.]
(b) Determine the marginal PMFs $p_X[i]$ and $p_Y[j]$. [5 pts.]
(c) Calculate $P[X \geq \frac{2}{\sqrt{2}}, Y \geq \frac{2}{\sqrt{2}}]$. [5 pts.]
(d) Compute $E[Y|X = 1]$. [10 pts.]

2. The continuous random variables X and Y have the joint probability density function (PDF)

$$p_{X,Y}(x,y) = k(x+y), \qquad 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$$

(a) Find k. [5 pts.]
(b) Calculate the marginal PDFs $p_X(x)$ and $p_Y(y)$. [10 pts.]
(c) Compute $P[X + Y > 0.5]$. [10 pts.]

3. Find the covariance of X and Y if they have the following joint PDF. [25 pts.]

$$p_{X,Y},(x,y) = e^{-(x+|y|)}, \qquad x > 0, \; -x < y < x$$

4. The continuous random variables X and Y have the following joint PDF

$$p_{X,Y},(x,y) = e^{-(x+y)}, \qquad 0 < y < x < 1$$

Let $W = X + Y$, $Z = X - Y$. Find $p_{W,Z}(w,z)$, the joint PDF of W and Z. [20 pts.]

Directions:

- One A4-sized formula sheet (without solutions) can be used.
- The exam is 80 minutes.

GOOD LUCK!

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1. (a)

θ	X	Y
0	2	0
1	$\sqrt{2}$	$\sqrt{2}$
2	0	2
3	$-\sqrt{2}$	$\sqrt{2}$
4	-2	0
5	$-\sqrt{2}$	$-\sqrt{2}$
6	0	-2
7	$\sqrt{2}$	$-\sqrt{2}$

(5 pts)

$\Rightarrow$

X ↓ / Y →	-2	$-\sqrt{2}$	0	$\sqrt{2}$	2
-2	0	0	1/8	0	0
$-\sqrt{2}$	0	1/8	0	1/8	0
0	1/8	0	0	0	1/8
$\sqrt{2}$	0	1/8	0	1/8	0
2	0	0	1/8	0	0

(5pts)

(b)

i	-2	$-\sqrt{2}$	0	$\sqrt{2}$	2
$P_X(i)$	1/8	2/8	2/8	2/8	1/8

(2.5 pts)

j	-2	$-\sqrt{2}$	0	$\sqrt{2}$	2
$P_Y(j)$	1/8	2/8	2/8	2/8	1/8

(2.5 pts)

(c) $P[X \geq \frac{2}{\sqrt{2}}, Y \geq \frac{2}{\sqrt{2}}] = P[X \geq \sqrt{2}] \cdot P[Y \geq \sqrt{2}]$
$= \frac{3}{8} \cdot \frac{3}{8} = \frac{9}{64}$ (5 pts)

d) First answer $E[Y|X=1] = E[Y] = (-2)\cdot\frac{1}{8} + (-\sqrt{2})\frac{2}{8} + 0\cdot\frac{2}{8} + \sqrt{2}\cdot\frac{2}{8} + 2\cdot\frac{1}{8} = 0$ since X and Y are independent

Second answer $E[Y|X=1]$ is not defined since $P[X=1]=0$. (10 pts)

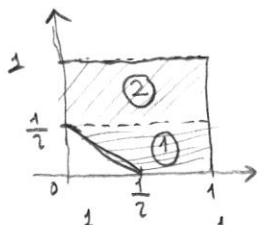
Both answers are acceptable!

2. (a) $\int_0^1\int_0^1 k(x+y)\,dy\,dx = k\int_0^1\left[\int_0^1 (x+y)\,dy\right]dx = k\int_0^1 \left(xy+\frac{y^2}{2}\Big|_0^1\right)dx = k\int_0^1 (x+\frac{1}{2})\,dx = k\cdot\left(\frac{x^2}{2}+\frac{1}{2}x\Big|_0^1\right)$ (5pts)

$= k\cdot\left(\frac{1}{2}+\frac{1}{2}\right) = k\cdot 1 = 1 \Rightarrow \boxed{k=1}$

(b) $P_X(x) = \int_0^1 (x+y)\,dy = xy+\frac{y^2}{2}\Big|_0^1 = x+\frac{1}{2}$ (5pts)

$P_Y(y) = \int_0^1 (x+y)\,dx = \frac{x^2}{2}+yx\Big|_0^1 = y+\frac{1}{2}$ (5pts)



(c) $P[X+Y>0.5] = \iint_{(x,y)\in①}(x+y)\,dx\,dy + \iint_{(x,y)\in②}(x+y)\,dx\,dy$

$= \int_0^{\frac{1}{2}}\left[\int_y^1 (x+y)\,dx\right]dy + \int_{\frac{1}{2}}^1\left[\int_0^1 (x+y)\,dx\right]dy = \int_0^1 \left(\frac{x^2}{2}+yx\Big|_y^1\right)dy + \int_{\frac{1}{2}}^1 \left(\frac{x^2}{2}+yx\Big|_0^1\right)dy$ (5pts)

(5pts)

$= \int_0^{\frac{1}{2}}\left(\frac{1}{2}y+y-\frac{3y^2}{2}\right)dy + \int_{1/2}^1 \left(\frac{1}{2}+y\right)dy = \left(y^2+\frac{y^2}{2}-\frac{y^3}{2}\Big|_0^{1/2}\right) + \left(\frac{1}{2}y+\frac{y^2}{2}\Big|_{1/2}^1\right) = \frac{5}{16}+\frac{5}{8} = \frac{15}{16}$

3. $E[XY] = \int_0^\infty\int_{-x}^{x} xy\,e^{-(x+|y|)}\,dy\,dx = \int_0^\infty x\,e^{-x}\left[\int_{-x}^{x} y\,e^{-|y|}\,dy\right]dx = \int_0^\infty x\,e^{-x}\left[\int_{-x}^{0} y\,e^{y}\,dy + \int_0^x y\,e^{-y}\,dy\right]dx$

(5pts)

$= \int_0^\infty x\,e^{-x}\left[(y-1)e^y\Big|_{-x}^0 - (y+1)e^{-y}\Big|_0^x\right]dx = \int_0^\infty x\,e^{-x}\left[-1+ye^{-y}+e^{-y}-ye^{-y}-e^{-y}+1\right]dx = \int_0^\infty x\,e^{-x}\,0\,dx = 0$

(5pts) $P_X(x) = \int_{-x}^{x} e^{-(x+|y|)}\,dy = e^{-x}\left[\int_{-x}^{0} e^{y}\,dy + \int_0^x e^{-y}\,dy\right] = e^{-x}\left[1-e^{-x}+1-e^{-x}\right] = 2e^{-x}(1-e^{-x}) = 2(e^{-x}-e^{-2x})$

(5pts) $P_Y(y) = \int_0^\infty e^{-(x+|y|)}\,dx = e^{-|y|}\cdot\int_0^\infty e^{-x}\,dx = e^{-|y|}\cdot\left(-e^{-x}\Big|_0^\infty\right) = e^{-|y|}$.

(5pts) $E[y] = \int_{-x}^{x} y\,P_Y(y)\,dy = \int_{-x}^{x} y\,e^{-|y|}\,dy = \int_{-x}^{0} y\,e^{y}\,dy + \int_0^x y\,e^{-y}\,dy = 0$ (calculated above!)

$E[x] = \int_0^\infty x\,P_X(x)\,dx = \int_0^\infty x\,2(e^{-x}-e^{-2x})\,dx$ (no need to compute since $E[Y]=0$.

(5pts) $Cov(x,y) = E[xy] - E[x]E[y] = 0 - 0 = 0$ (X and Y

4. $X = \frac{W+Z}{2}$, $Y = \frac{W-Z}{2}$ (5pts)

$A = \begin{bmatrix} \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \\ \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $|\det(A)| = |-2| = 2$ (5pts)

$P_{W,Z}(w,z) = \frac{1}{2}P_{X,Y}\left(\frac{w+z}{2}, \frac{w-z}{2}\right)$ (5pts)

$= \frac{1}{2}e^{-\left(\frac{w+z}{2}+\frac{w-z}{2}\right)} = \frac{1}{2}e^{-w}$ (5pts)